CATERPILLAR®

DEALER BEST PRACTICE SERIES

Improving GET Replacement Process

Maintenance and Repair | **Application** | Component Life Management | Component Renewal (CRC) | MARC Management

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1.0 Introduction

This Best Practice provides faster and better logistics, plus increases safety of Ground Engagement Tools (GET) replacement and repairing on the job site. Additionally, it supports efficient repair and transport of the components and GET tools, and the service performance in field such as welding and washing.

2.0 Best Practice Description

Marcosa built a GET truck to better meet the customer's needs.

Features of the truck include:

- Platform box for parts storage
- Cabinets for storage
- Generator – 25Kva three phase electrical power
- Air compressor
- Welding machine
- Work Room with counter tops and vise
- Water pressure vessel with manual retraction reel
- Crane
- Rear platform box to support storage and transport of GET

A detailed description of the truck features can be found in Section 6.0.

GET Replacement:

The GET truck attends the field repairing and replacing GETs and undercarriage, transporting and storing the backup parts. Note: A tag line should be used to control the load.

Support for the corrective and inspection team:

- Repairing of undercarriage (re-torque bolts)
- Utilization of the truck (crane) to move components
- Welding Services
- Blowtorch services
- Support with the High Torque (compressed air system)
- Washing of machines and components with high pressured water
- Utilization of GET tools (components assembly bench, etc.)
- Energy providing 110v ~220v




Support for the main workshop:

- Removal of components
- Removal of discarded material.
- Transporting of components on the jobsite.

3.0 Implementation Steps

The process started with the truck. Choosing the best truck and adapting it to the customer's need, then listing the items and designing the tools and features. See Section 6.0 for truck features description.

The date of implementation was May 15, 2010. It took one and a half months to build and implement. People involved were: Contract manager, Field Supervisor, technicians, and CI Champion.

The project was laid out in the following 6 Sigma format (Non 6 Sigma Project):

- Define: Design the truck and features
- Measure: Investment and utilization
- Analyze: Benefits
- Improve: Redesign – Change/Add features and details
- Control: Measure improvement / utilization

4.0 Benefits

The benefits of the truck include:

- Provides increased safety
- Improves the logistic process
- Provides faster and more efficient replacement and repair of GETs and components
- Providing better support on the jobsite with complementary services such as: welding, washing and transportation of GET- parts and components
- Transport of new, scrap, and GET needing repair
- Improve quality

Financial benefits for Marcosa have included:

- Reduced downtime for the customer because the service is performed in the field, resulting in increased availability.
- Marcosa receives a bonus of up to 2.5% of the value of the contract by increasing the availability over target.

5.0 Resources Required

The truck created by Marcosa consisted of the following cost. Truck and modification details can be found in Section 6.0.

- 01 Truck (Volkswagen 13.180 Constellation Standard – $80,760 USD (BRL$: 136.000,00))
- Modifications for GET Support (See Section 6.0 for details – $30,285 USD (BRL$ 51.000,00))
- 01 Munck (Crane) (Model MK 6,5T – $23,705 USD (BRL$ 39.980,00))

Total Investment: $134,750 USD **(BRL$ 250.980,00)**

6.0 Supporting Attachments / References

Base Truck Information

Engine:
MWM 4.12 TCE 4-cylinder - Turbo and intercooler - Common Rail. Power - 180 hp Euro III / 132 kw / 2200 rpm Torque - 600 Nm / 61 mkgf / 1550 to 2000 rpm Alternator - 80A - 14V Battery - 2 X (12V - 100 Ah) 24V Injection system: Common Rail

Transmission:
Box Models - Eaton FS 4205 A
Gears - 5 forward, 1 reverse - 5 speed
Clutch - Sachs, dry disc
Drive - push type self-tuning

Cabin:
Constellation Stand

Chassis:
Type - U LNE Profile Ladder 38 (cm3) 163
Tubeless radial tires - 900R X 20-14 PR
Steering - ZF 8095 - Hydraulic
Plastic Fuel Tank - 275 liters

Weight (kg):
4,340 = 8,375 / 4,800 = 8,315 (kg) / 5207 = 8255 (Kg)

Truck Modifications for GET Support

CJ.	QUANTITY	COMPOSITION
STRUCTURE	1	**Platform** (dry cargo) 2000 mm long, geometrically designed to absorb all kinds of active efforts. Constructed in carbon steel plate, SAE1010/20, joined by electric arc type MIG.
	1	**Side Guard,** fabricada made from carbon steel plate with SAE1010/20 450mm high, hinges and lock nuts (butterfly type) for closing.
	1	**Step back**, in carbon steel plate SAE1010/20 slip.
	2	**Rear cabinets,** Made by folded profiles and panels on carbon steel plates SAE 1010/20, with doors, special locks and sealing through automotive rubber against dust and water. The doors have access controlled by a key lock. Doors with vertical opening are supported by gas shock absorbers.
	1	**Duralumin Trunk polished and creased,** with thermal insulation on the ceiling, internal galvanized iron rods, chess carbon steel plate, thickness of 3/16”, one left side door (driver side), one front window and one central window on the right side.
	1	**Attached to the vehicle,** made through metal plates on the rear, preventing the longitudinal displacement of the implement, guides and shock absorbing system with elastomer (plastprene) on the front.
	2	**Rear fender,** made of carbon steel plate, with a transversal 1/8" plate, and 2mm folded frame.
	2	**Rear anti-clay,** made of synthetic rubber with the IMPACTO logo.
	1	**Rear bumper,** articulated, in accordance to the norm RTQ32/04 of INMETRO, resolution 152/03 of CONTRAN and decree 11/04 of. DENATRAN

FUNCIT ION		**Trigger set,** mechanical, through the PTO/TDF properly sized, with reinforced driveshaft, bearings, base, mounting brackets and strap stretching system.

		Micro-adjustable throttle for rotation control.

GENERATOR	1	**Generator 25Kva three phase eletrical Power,** auto-excited and auto-regulated by compensating system based on winding excitation, Electromagnetic circuit built by discs printed on laminated silicon steel or treated low carbon steel, enameled wires of class H (180 °) and insulating quality. Housing in cold rolled steel welded with anticorrosive treatment. Insulation Class F (155 ° C)- Protection class IP21Norma construction and testing: ABNT / IEC / NEMA. Powered by hydraulic actuator via the control lever
	1	**Protection panel and reading of instruments,** assembled on closed steel Box for vertical standing, containing ammeter, voltmeter, frequency meter and breaker.

AIR	1	**Air compressor Piston**, two stages in line with displacement 20PCM nominal maximum pressure of 175 Kg, powered by three phase 220V electric motor and equipped with pressure switch, safety valve and records, prepared to work intermittently and the air tank of 200 liters.
	1	**Dehumidifier air coalescing filter** with retention of 99% humidity.
	1	**Manual retraction reel** with 20 feet of hose 1 / 4 "quick coupler, and hose connection.

WELDING	1	Welding machine model TDC435 ED with three-phase power source, Digital Electronic Technology DSP (Digital Signal Processor) for welding with coated electrodes and TIG process in direct current (DC). Facility for the operator, which allows you to adjust the welding current with the machine turned on empty, thereby preserving the values on the display, after welding Control range from 500 to 400 A. Gauge electrode 6010/6011/6013/7018 until 6mm.

WATER	1	Pressure vessel with a capacity of 200 liters, built in carbon steel SAE 1010/1020, with spherical heads, flange for cleaning and inspection, filler type Kanlock, venture suction pump vacuum type, safety valve with regulator pressure gauge with full scale of 300psi, suction hose 3 (three) meters long, a record sphere type size 1” for controlling, level display (maximum and minimum), Handbook MANUFACTURING AS STANDARD NR-13. Lower output sphere type “1. Lower tank ready for discharge lubricant and also self loading, made of air. "Does not require booster pump, "(patent pending).
	1	**Manual retraction reel**, hose Ø1 / 2 "with 10 meters in length and adjustable nozzle.

ACCESSORIES	1	**Bench vise nº8, installed on the services bench.**
	1	**Motoesmeril industrial,** motor elétrico de 1/2cv monofásico, 60Hz, 220 volts com rebolo de 6” x ¾” x ½”.
	1	Services Bench dimensions of 1200mm (length) x 700mm (width) x 1000mm (height), with two drawers, two doors and locks, sealed through automotive rubber. Constructed in carbon steel

		plate, thickness of 2mm and top made of wood.
	1	**Oxy-acetylene set,** with support for accommodation of cylinders, built in carbon steel plate based on 3 / 16 " and body 1 / 8". Support for wrapping hoses, 5m double hose for blowtorch, pen to cut, cutting flame valve and pressure regulators. (Cylinders not included with the set);
	1	**Internal cabinet for tools,** dimensions 1000mm (length) x 500mm (width) x 1600mm (height), built entirely of carbon steel plate of 2mm, with two divisions for the storage of tools and four doors with locks with automotive rubber for sealing.
	1	**Reversing alarm**
	1	**Spare tire**
	2	**Support with fire extinguisher of 8kg - Dry Chemical**
	1	**Plastic reservoir for drinking water with capacity of 20 liters**

7.0 Related Best Practices

Best Practice Number: 1006-1.05-1208, Backhoe-Hammer to remove large mining GET
Best Practice Number: 1010-2.06-1214, Auxiliary Fleet and Loaders GET Chart for Preventive Maintenance

8.0 Acknowledgements

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